DS-3002 Data Mining Assignment 2 Report

BSDS — FAST-NUCES — Spring 2026

March 28, 2026

Scenario

A UK online gift retailer asked for a customer intelligence pipeline to uncover:

- Co-purchase structure (FP-Growth),
- Segment-level behavior (CARs with RFM labels),
- Ordered journey patterns and churn-warning triggers (Sequential Mining).

Dataset

- UCI Online Retail II: `online_retail_II.xlsx`
- Sheets used: Year 2009–2010 and Year 2010–2011
- Retail line-item transactions with invoice, stock code, quantity, time, and customer ID

1 Required Preprocessing

Pipeline Steps

1. Load and concatenate both sheets.
2. Remove cancellations (Invoice starts with C), null customer IDs, and non-positive quantity/price.
3. Restrict to United Kingdom.
4. Build invoice-level transaction sets for FP-Growth and CARs.
5. Build customer-level chronological event sequences for sequential mining.
6. Report key post-cleaning metrics.

Post-Cleaning Summary

Metric	Value
Invoices	33,541
Customers	5,350
Products	4,616
Date range	2009-12-01 07:45:00 to 2011-12-09 12:49:00
Average basket size	20.57 items/invoice

2 Part A: FP-Growth – Efficient Co-Purchase Mining

2.1 A1. FP-Tree Walkthrough

A six-invoice sample was used to manually illustrate FP-Tree construction: item frequency counting, descending-order insertion, shared-prefix compression, and conditional pattern base intuition.

2.2 A2. Full FP-Growth and Rule Quality

Configuration: minsup = 2%, minconf = 50%.

Itemset Size	Count
1-itemsets	193
2-itemsets	28
3-itemsets	0
Total	221

Table 1: Frequent itemsets from FP-Growth

Top-15 rules were ranked by lift in the notebook and used for bundle design.

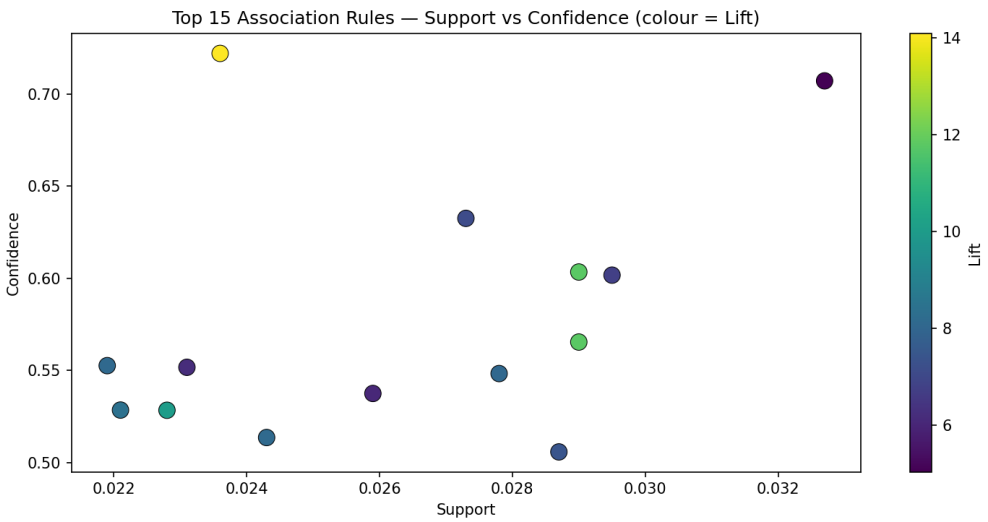


Figure 1: Association rule landscape: support vs confidence (color = lift)

2.3 A3. Product Bundle Recommendations

Two concrete gift bundles were proposed from high-confidence/high-lift rules (shown in notebook output tables), with business rationale based on strong co-purchase affinity.

3 Part B: Class Association Rules – Customer Segmentation

RFM Labeling

Quartile-based thresholds derived from the notebook output:

- Recency: $Q1 = 25$ days, $Q3 = 381$ days
- Frequency: $Q1 = 1$, $Q3 = 7$
- Monetary: $Q1 = 336.17$, $Q3 = 2214.91$

Segment counts:

- At-Risk: 3458
- One-and-Done: 1272
- Champion: 620

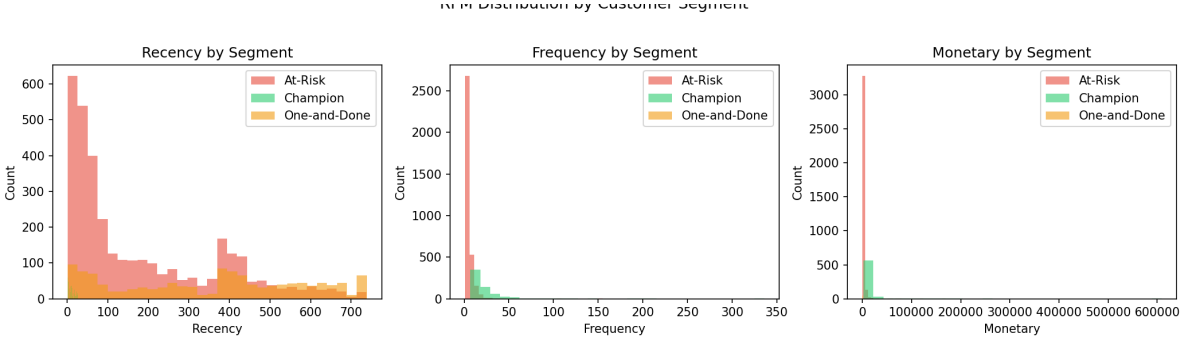


Figure 2: RFM distribution by segment

3.1 B1. CAR Mining per Segment

With minsup = 1% and minconf = 55%, mined rule counts were:

Segment	CAR Count
Champion	620
At-Risk	3458
One-and-Done	1272

3.2 B2. New Customer Prediction

For sample items {82482, 22698, 22579}, rule firing showed evidence across all three segments, with highest confidence in One-and-Done (0.8889) and strong At-Risk/Champion alternatives as secondary signals.

4 Part C: Sequential Pattern Mining – Customer Journey

Category Mapping

Keyword-based mapping over Description created coherent categories: Candles & Home Fragrance, Seasonal & Festive, Kitchen & Dining, Storage & Organisation, Stationery & Gifts, Garden & Outdoor, Lighting & Decor, and Other.

Coverage from notebook output: 70.8% transactions mapped to named categories (non-Other).

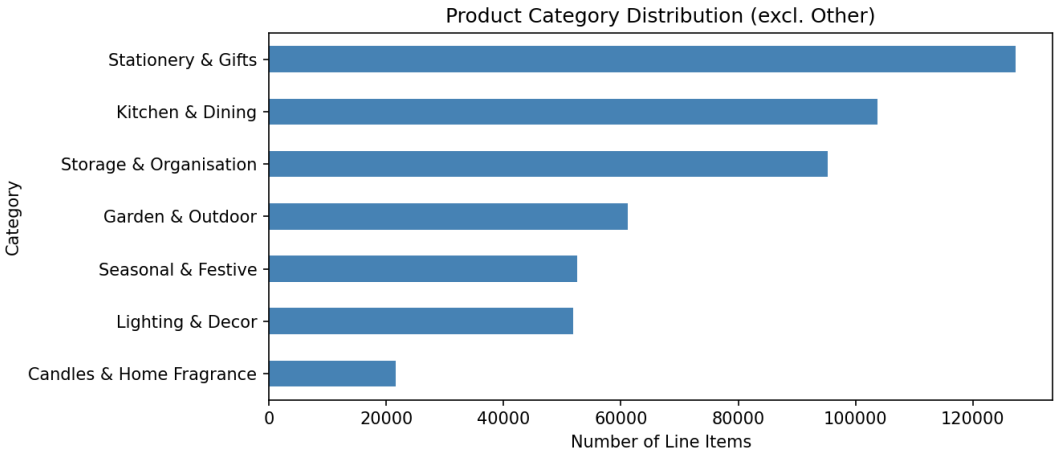


Figure 3: Category distribution (excluding Other)

4.1 C1. GSP-family Mining (PrefixSpan)

- Total customer sequences: 5350
- Average sequence length: 6.27 invoices/customer
- minsup (absolute): 26
- Total frequent sequences: 3933
- Length-1: 141, Length-2: 925, Length-3: 1017

4.2 C2. At-Risk Subpopulation Mining

At-risk-only sequence mining was performed on the segmented customer subset and produced:

- Length-2 at-risk sequences: 391
- Length-3 at-risk sequences: 169

4.3 C3. Early-Warning Trigger Rules

Three trigger candidates were produced from top at-risk length-2 patterns:

Trigger	Support	Confidence	Estimated flagged/month
T1	348	76.82%	14.15
T2	263	35.30%	10.69
T3	256	56.51%	10.41

Table 2: Early-warning churn triggers from at-risk sequences

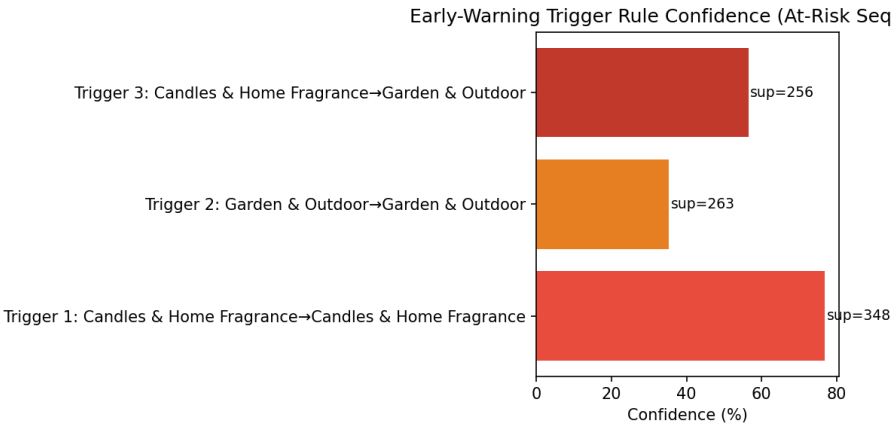


Figure 4: Trigger rule confidence comparison

Conclusion

The assignment pipeline was completed end-to-end from one notebook run: preprocessing, co-purchase mining, segment-aware rule mining, and actionable churn triggers. The outputs support both merchandising decisions (bundles) and retention operations (early-warning discounts).